

Robust Risk Under Evolving Uncertainty: A Wasserstein Counterpart of the Entropic Value-at-Risk

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Abstract

An agent still learning its environment should be cautious while ignorant and bold once confident. The entropic value-at-risk captures this through a robust-optimization identity—a confidence level fixes the radius of a relative-entropy ball of alternative models—but that ball cannot reach catastrophes the nominal deems impossible, precisely what a safe agent must hedge. We instead use an optimal-transport ball and study the coherent risk measure it induces, the Wasserstein entropic value-at-risk. It has a variational dual mirroring the entropic formula (an inverse temperature becomes a transport price), occupies a definite place in the risk hierarchy, and provably accounts for the reachable catastrophes the entropic measure ignores; we verify both dualities numerically. Driving the transport radius by belief entropy then yields a closed-form robust dynamic-programming operator whose caution contracts as the belief sharpens, with a certified safety sandwich and a sharp safety switch.

1 RISK UNDER EVOLVING UNCERTAINTY

Consider a drone crossing a canyon under uncertain wind. A *high-cruise* policy is fast when the air is calm but is thrown into the cliffs by a sudden gust; a *hover-and-creep* policy is safe in any wind but slow. The wind regime—a hidden type z drawn once—is never observed directly; the drone sees only noisy signals and must infer z as it flies. Two reflexes each fail. A controller that assumes the *worst case forever* (robust control [Iyengar, 2005, Nilim and El Ghaoui, 2005, Wiesemann et al., 2013]) keeps hovering even after

the air has visibly calmed: safe, but so conservative that operators switch it off. A controller that *commits before its belief sharpens* (expected-value or Bayesian planning [Kaelbling et al., 1998, Ghavamzadeh et al., 2015]) cruises on a hunch, and a single wrong guess about z is fatal—an expected return that blends a benign mode with a catastrophic one is a number nobody actually attains. What is wanted is a controller whose caution *evolves with its evidence*—maximal under ignorance, vanishing once the environment is identified—together with a safety bound that holds at *every* stage of learning. The agent’s risk attitude must therefore be a function of its *evolving* epistemic state, and the entropy of its belief is a live readout of how much it should distrust its own model.

The entropic value-at-risk [Ahmadi-Javid, 2012] turns this intuition into algebra. It is the unique single-parameter coherent family that sweeps from the mean to the essential supremum, and it has an exact *distributionally robust* representation: its confidence level α equals the radius $-\ln \alpha$ of a relative-entropy ball of alternative laws. Ganguly et al. [2025] further show the induced optimization is convex and admits well-posed, convergent estimation. This robust reading is what we want for evolving uncertainty: “how conservative should I be?” becomes “how large should my ambiguity set be?”, and the latter is answered by the agent’s own uncertainty.

But the relative-entropy geometry has a blind spot that is decisive for *safety*. A ball $\{Q : \text{KL}(Q\|P) \leq \rho\}$ contains only laws absolutely continuous with respect to the nominal P : if P assigns zero probability to a catastrophic transition, no member assigns it positive probability, since KL is infinite off the support of P . As the agent learns and its nominal kernel concentrates away from rare disasters, the relative-entropy adversary loses the ability even to *represent* those disasters, so the safety check becomes *vacuous* precisely when overconfidence is most dangerous. The optimal-transport (Wasserstein) geometry has no such blind spot: it prices a perturbation by how far mass must physically move, so a verifier can still ask “what if it is a storm?” at a cost

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proportional to the distance to that outcome—*reachable but expensive* [Mohajerin Esfahani and Kuhn, 2018, Gao and Kleywegt, 2023, Blanchet and Murthy, 2019].

Related work. Risk-averse dynamic programming [Ruszczynski, 2010], risk-sensitive control and reinforcement learning [Howard and Matheson, 1972, Chow et al., 2015, Hau et al., 2023], percentile and parameter-uncertainty MDPs [Delage and Mannor, 2010], constrained and safe RL [Altman, 1999, García and Fernández, 2015, Sui et al., 2015, Brunke et al., 2022], and adversarially- or Wasserstein-robust RL [Pinto et al., 2017, Abdullah et al., 2019] all add caution to sequential decisions—but with a *fixed* attitude. Our contribution is to let the ambiguity radius, hence the risk attitude, be read off the agent’s evolving belief, and to give it the optimal-transport geometry safety requires [Kuhn et al., 2019].

Contributions. We keep the entropic value-at-risk’s robust formulation and its guarantees, and replace its ball.

- We define the **Wasserstein entropic value-at-risk** (WEVaR) and prove a Kantorovich–Rubinstein **variational dual** (Thm. 1) that mirrors the entropic formula term for term, is **convex and well-posed** (the transport analogue of the guarantee of Ganguly et al., 2025), and has a closed “mean-plus-Lipschitz” form.
- We show it is **coherent** and place it in the risk hierarchy (Thm. 2): it is sandwiched against the entropic measure by a transport–entropy inequality and **strictly accounts** for zero-nominal-probability catastrophes the entropic measure ignores.
- We **verify** both robust-optimization problems, their dualities, and the comparison numerically with Gurobi (§4), confirming strong duality to solver tolerance.
- Driving the transport radius by **belief entropy** gives a closed-form robust dynamic-programming operator (Thm. 3) and a computable safety switch under evolving belief (§5).

2 THE ENTROPIC VALUE-AT-RISK AND ITS ROBUST FORMULATION

Let X be a bounded loss on a finite space $\mathcal{S}$ with reference law P , and let $d : \mathcal{S} \times \mathcal{S} \rightarrow \mathbb{R}_{\geq 0}$ be a ground metric with diameter $D = \max_{s,s'} d(s, s')$. A risk measure is *coherent* [Artzner et al., 1999] if it is monotone, translation-invariant, positively homogeneous and subadditive; the conditional value-at-risk [Rockafellar and Uryasev, 2000] and the entropic value-at-risk are its canonical instances. The latter has the dual/primal pair

$$\text{EVaR}_\alpha(X) = \inf_{t>0} \frac{1}{t} (\ln \mathbb{E}_P[e^{tX}] - \ln \alpha) \quad (1)$$

$$= \sup \{ \mathbb{E}_Q[X] : \text{KL}(Q\|P) \leq -\ln \alpha \}, \quad (2)$$

where the inverse temperature t in (1) is the multiplier on the relative-entropy constraint in (2), and the worst-case law is the exponential tilt $Q_s^* \propto P_s e^{t^* X_s}$ [Ahmadi-Javid, 2012, Ganguly et al., 2025]. Two facts we carry forward from Ganguly et al. [2025]: (G1) the reparametrized objective is convex, so (1) is a well-posed one-dimensional convex program with a unique solution; (G2) the tilt Q^* is supported on $\text{supp}(P)$. Property (G2) is the blind spot: $\text{EVaR}_\alpha(X)$ is independent of the value of X on any zero-nominal-probability state, *for every* α .

3 SWAPPING THE BALL: A WASSERSTEIN ROBUST RISK

We keep the robust template (2) and replace the relative-entropy ball by a 1-Wasserstein ball, $\mathcal{W}_1(Q, P) = \min_{\gamma \in \Gamma(Q, P)} \sum_{s,s'} d(s, s') \gamma(s, s')$.

Definition 1 (WEVaR). For radius $\varepsilon \geq 0$,

$$\text{WEVaR}_\varepsilon(X) = \sup \{ \mathbb{E}_Q[X] : \mathcal{W}_1(Q, P) \leq \varepsilon \}. \quad (3)$$

Theorem 1 (Variational dual, convexity, closed form). *For bounded X and $\varepsilon \geq 0$,*

$$\text{WEVaR}_\varepsilon(X) = \inf_{\lambda \geq 0} \{ \lambda \varepsilon + \mathbb{E}_P[X_\lambda^c] \}, \quad (4)$$

where $X_\lambda^c(s) = \max_{s'} (X(s') - \lambda d(s', s))$ is the c -transform of X . The objective is convex in λ , so (4) is a well-posed one-dimensional convex program (mirroring (G1)); the optimal price satisfies $\lambda^ \leq \text{Lip}_d(X)$; $\varepsilon \mapsto \text{WEVaR}_\varepsilon(X)$ is concave and nondecreasing; and*

$$\text{WEVaR}_\varepsilon(X) = \mathbb{E}_P[X] + \varepsilon \text{Lip}_d(X) \quad (5)$$

for ε below a saturation threshold (in particular, exactly for two-point supports), where $\text{Lip}_d(X) = \max_{s \neq s'} |X(s) - X(s')|/d(s, s')$.

Proof sketch. Strong duality for Wasserstein DRO [Gao and Kleywegt, 2023, Blanchet and Murthy, 2019] gives (4); the c -transform X_λ^c is the transport counterpart of the cumulant $\frac{1}{t} \ln \mathbb{E}_P e^{tX}$ in (1). Convexity in λ holds because X_λ^c is a pointwise maximum of affine functions of λ , hence convex, and $\mathbb{E}_P$ and the $\lambda \varepsilon$ term preserve convexity. For $\lambda \geq \text{Lip}_d(X)$ the maximum is attained at $s' = s$, so the bracket equals $\mathbb{E}_P[X]$ and the objective is $\mathbb{E}_P[X] + \lambda \varepsilon$, minimized at $\lambda = \text{Lip}_d(X)$; this gives (5) until transporting all displaceable mass to the maximizer of X saturates the bound, after which the value rises concavely toward $\max_s X(s)$. $\square$

Equations (1) and (4) are the same template—inf over a one-dimensional dual variable of a *smoothed* expectation plus radius-times-multiplier—with an entropic smoothing for the relative-entropy ball and a transport (inf-convolution)

smoothing for the Wasserstein ball. The temperature t and the transport price λ are the same Lagrangian object on two ambiguity geometries.

Proposition 1 (Coherence). *WEVaR $_\varepsilon$ is a coherent risk measure for every $\varepsilon \geq 0$.*

Proof sketch. It is the support function of the convex, compact set $\{Q : \mathcal{W}_1(Q, P) \leq \varepsilon\}$, hence positively homogeneous and subadditive; P lies in the set, giving monotonicity; and adding a constant to X shifts every $\mathbb{E}_Q[X]$ equally, giving translation invariance. $\square$

Theorem 2 (Hierarchy and relation to the entropic measure). *On a finite metric space with diameter D : (i) $\text{WEVaR}_0(X) = \mathbb{E}_P[X]$ and $\text{WEVaR}_\varepsilon(X) \uparrow \text{ess sup}(X)$ as $\varepsilon \uparrow D$, so WEVaR sweeps the full hierarchy; (ii) (sandwich) $\text{EVaR}_\alpha(X) \leq \text{WEVaR}_{D\sqrt{-\frac{1}{2}\ln \alpha}}(X)$ for all $\alpha \in (0, 1]$; (iii) (catastrophe) if $P(s^*) = 0$ and $X(s^*) > \mathbb{E}_P[X]$, then $\text{EVaR}_\alpha(X)$ is independent of $X(s^*)$ for every α , whereas $\text{WEVaR}_\varepsilon(X)$ strictly increases in $X(s^*)$ once $\varepsilon > \text{dist}_d(s^*, \text{supp } P)$.*

Proof sketch. (i) is immediate from finiteness. (ii): Pinsker gives total variation $\leq \sqrt{\text{KL}/2}$ and $\mathcal{W}_1 \leq D \cdot \text{TV}$, so the relative-entropy ball of radius $-\ln \alpha$ is contained in the Wasserstein ball of radius $D\sqrt{-\frac{1}{2}\ln \alpha}$; take suprema [Bobkov and Götze, 1999, Boucheron et al., 2013]. (iii) is (G2) versus the transport reach: no finite-radius relative-entropy ball contains a Wasserstein ball, and the gap is exactly the reachable-but-zero-probability catastrophes. $\square$

4 EMPIRICAL GUARANTEES

We verify both robust programs and their duals with Gurobi 12 on a five-state metric space ($X = [0, 2, 4, 8, 100]$, $d(s, s') = |s - s'|$). For the entropic measure we solve the relative-entropy-constrained program (2) (a nonlinear program) and match it against the convex primal (1); for WEVaR we solve the transport linear program (3) and match it against the Kantorovich–Rubinstein dual (4).

Strong duality holds to solver tolerance: the entropic primal (1) and the relative-entropy program (2) agree to 1.2×10^{-4} , and the transport linear program agrees with its Kantorovich–Rubinstein dual to 9×10^{-7} . The closed form (5) is exact in the unsaturated regime (here $\varepsilon \leq 0.1$) and is otherwise upper-bounded by the exact one-dimensional dual, which the solver confirms. Across a sweep of radii both measures rise from the mean toward the worst case, with $\text{EVaR}_\alpha \leq \text{WEVaR}$ at the Pinsker-matched radius (Thm. 2(ii)). The catastrophe blind spot is decisive: with a zero-nominal-probability disaster whose loss we grow from 50 to 1000, the entropic value-at-risk changes by exactly 0

KL ball (red —) is trapped on the support edge;
 $\mathcal{W}_1$ ball (blue) reaches the Fail vertex

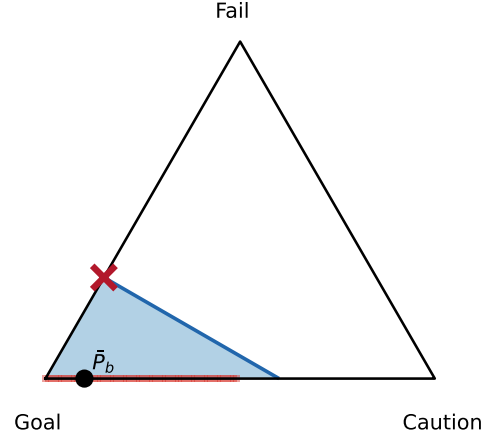


Figure 1: Why Wasserstein, geometrically. On the next-state simplex at a confident belief—the nominal $\bar{P}_b$ places zero mass on FAIL—the relative-entropy ball (red) is trapped on the support edge, blind to the catastrophe, while the 1-Wasserstein ball (blue) reaches the FAIL vertex; the worst-case adversary ($\times$) sits on its boundary. This is Thm. 2(iii) drawn on $\Delta(S)$.

at fixed confidence, while WEVaR at a fixed radius changes by 807.5—the gap of Thm. 2(iii), shown geometrically in Fig. 1 and quantitatively across radii in Fig. 4 (App. E).

5 EVOLVING UNCERTAINTY: BELIEF ENTROPY AS THE RADIUS

We now let the radius *evolve*. An environment has a hidden type $z \in \mathcal{Z}$ drawn once; the agent maintains a Bayesian belief $b \in \Delta(\mathcal{Z})$ with update ψ , forms the belief-weighted nominal kernel $\bar{P}_b(\cdot | s, a) = \sum_z b(z)P(\cdot | s, a, z)$, and sets the transport radius to the belief entropy, $\varepsilon(b) = \beta \mathcal{H}(b)$ with $\mathcal{H}(b) = -\sum_z b(z) \ln b(z)$ and sensitivity $\beta > 0$, giving the ambiguity set $\mathcal{U}(b) = \{Q : \mathcal{W}_1(Q, \bar{P}_b) \leq \beta \mathcal{H}(b)\}$. As observations sharpen b , the entropy $\mathcal{H}(b) \downarrow 0$, the ball contracts and drifts toward the nominal (Fig. 2), and via Thm. 1 the agent’s risk attitude slides from worst-case to risk-neutral—a dial driven by epistemic state. Since the controller minimizes value, the relevant functional is the infimal twin of Def. 1, whose closed form is $\mathbb{E}_{\bar{P}_b}[V] - \beta \mathcal{H}(b) \text{Lip}_d(V)$.

Theorem 3 (Closed-form robust update; contraction; safety). *For the operator, with discount $\gamma \in (0, 1]$ and belief update $b' = \psi(b, s, a, s')$,*

$$(\mathfrak{T}V)(s, b) = \max_a [r(s, a) + \gamma \inf_{Q \in \mathcal{U}(b)} \mathbb{E}_Q V(s', b')]$$

(i) *the inner problem has the closed form of Thm. 1—no coupling linear program—reducing the per- (s, a, b) cost to*

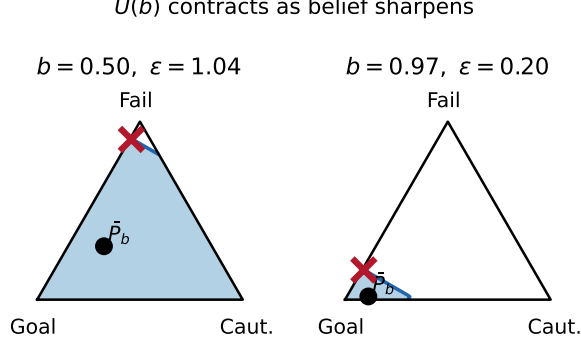


Figure 2: The entropy-modulated ambiguity set $\mathcal{U}(b)$ on the next-state simplex. As the belief sharpens the radius $\varepsilon = \beta \mathcal{H}(b)$ contracts ($1.04 \rightarrow 0.20$) and the nominal $\bar{P}_b$ drifts toward the goal; the worst-case adversary ($\times$) loses its reach toward the failure vertex—robustness that vanishes with uncertainty (the geometric content of Thm. 3).

$O(|\mathcal{S}|^2)$; (ii) $\mathfrak{T}$ is a γ -contraction in $\|\cdot\|_\infty$ when $\gamma < 1$, and a contraction on proper MDPs when $\gamma = 1$; either way it has a unique fixed point V^* [Bertsekas and Tsitsiklis, 1996]; (iii) V^* obeys a safety sandwich $V_{wc}^*(s, b) \leq V^*(s, b) \leq \mathbb{E}_{z \sim b}[V_{opt}^*(s, z)]$ between the always-maximally-cautious value and the type-aware oracle, and as $\mathcal{H}(b) \rightarrow 0$ the radius vanishes and $V^*(s, b) \rightarrow V_{opt}^*(s, z^*)$.

Since $\mathcal{H}(b) \leq \ln |\mathcal{Z}|$, the worst-case radius is $\beta \ln |\mathcal{Z}|$; requiring one action to remain viable under maximal ignorance gives the calibration $\beta < D / \ln |\mathcal{Z}|$, the transport analogue of choosing a confidence level.

Evolving safety switch. On the Ambiguous Bridge ($\gamma = 1$), SPRINT reaches the goal (+100) under a benign type but falls to catastrophe (−1000) under an adversarial type, while CRAWL is always safe but costly. With belief $b = [\alpha, 1 - \alpha]$ and $d(S_G, S_F) = 1$, so $\text{Lip}_d(V) = 1100$, Thm. 3(i) gives in closed form $Q(\text{SPRINT}) = -1 + 1100(\alpha - \beta \mathcal{H}(\alpha))^+ - 1000$ and $Q(\text{CRAWL}) = 80 - 1100 \min(\beta \mathcal{H}(\alpha), 1)$, which a Gurobi transport solve reproduces to 10^{-13} . Equating them (the entropy terms cancel) yields a safety switch independent of β : the agent CRAWLS until $\alpha \geq \alpha^* = 1081/1100 \approx 0.983$, then SPRINTS. The threshold is set purely by the reward asymmetry. The same creep-then-cruise switch governs the canyon of §1: the drone hovers until its confidence that the air is calm crosses ≈ 0.63 , then cruises (Fig. 6, App. E).

Guarantees, illustrated. On a five-state corridor whose forward action slips to a failure state only under a hidden “storm” type (Fig. 3), the safety sandwich of Thm. 3(iii) holds at every belief: the adaptive value never drops below the always-maximally-cautious floor and never exceeds the type-aware oracle. The price of robustness is *temporally*—the ceiling gap falls from 8.4 at the uniform belief

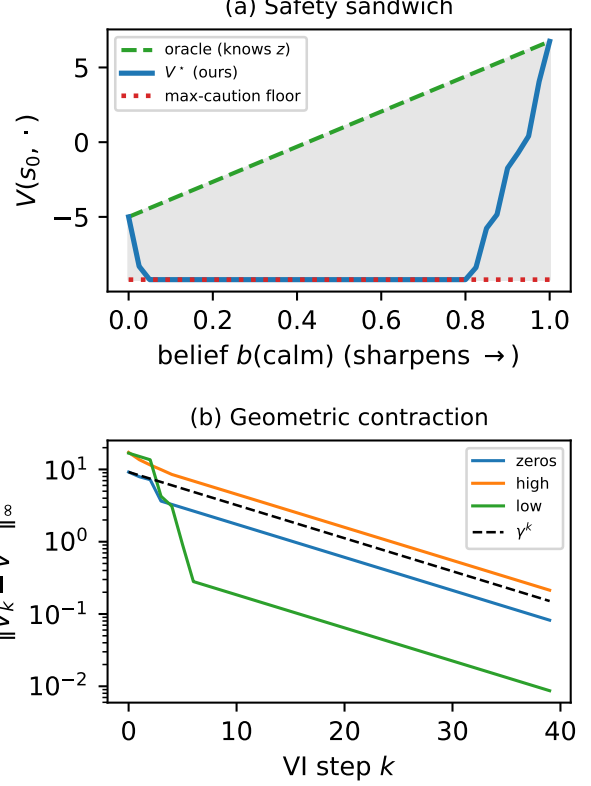


Figure 3: Computed guarantees on a five-state corridor with a hidden slip type ($\gamma = 0.9$, Gurobi-checked). (a) The adaptive value $V^*(s_0, b)$ stays inside the safety sandwich between the always-maximally-cautious floor V_{wc} and the type-aware oracle ceiling; the price of robustness shrinks as the belief sharpens. (b) From arbitrary initializations, value iteration with the closed-form operator collapses geometrically onto V^* along the γ^k envelope, certifying the bound at every iterate.

to 4.4 once the belief reaches $b(\text{calm}) = 0.95$, and to zero at identification. Value iteration with the closed-form operator converges from arbitrary initializations at empirical rate $0.90 = \gamma$, and the closed-form inner update matches a Gurobi transport solve at every sampled belief (the underlying duality is verified to solver tolerance in §4), so a valid safety bound is available at every iterate, not only at convergence. The belief-simplex manifolds (Fig. 5) and the safety-vs-efficiency rollouts (Fig. 7) appear in App. E.

6 DISCUSSION

The entropic and Wasserstein robust risks are the relative-entropy and optimal-transport faces of one idea—a coherent risk whose ambiguity radius is the agent’s epistemic uncertainty—bridged by entropic optimal transport [Cuturi, 2013, Peyré and Cuturi, 2019]; scaling the closed-form operator to continuous spaces via Lipschitz critics and to active information gathering are the natural next steps.

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APPENDIX: EXTENDED PROOFS

Throughout, $\mathcal{S}$ is finite with $|\mathcal{S}| = n$; d is a metric on $\mathcal{S}$ with diameter $D = \max_{s,s'} d(s, s')$; $P \in \Delta(\mathcal{S})$ is the reference law; $X : \mathcal{S} \rightarrow \mathbb{R}$ is a bounded loss; and $\mathcal{W}_1(Q, P) = \min_{\gamma \in \Gamma(Q, P)} \sum_{s,s'} d(s, s') \gamma(s, s')$ with $\Gamma(Q, P)$ the couplings of Q and P . We write $\mathbb{E}_Q[X] = \sum_s Q(s)X(s)$.

A. THEOREM 1

(a) Duality. Since $\mathcal{W}_1(Q, P)$ is itself a minimum over couplings, the worst-case expectation is the value of a single linear program over the coupling $\gamma \geq 0$ whose first marginal is fixed to P :

$$\text{WEVaR}_\varepsilon(X) = \max_{\gamma \geq 0} \sum_{s,s'} \gamma(s, s') X(s') \quad \text{s.t.} \quad \sum_{s'} \gamma(s, s') = P(s) \quad \forall s, \quad \sum_{s,s'} d(s, s') \gamma(s, s') \leq \varepsilon, \quad (6)$$

where the perturbed law is the second marginal $Q(s') = \sum_s \gamma(s, s')$ (automatically in $\Delta(\mathcal{S})$ because $\sum_{s,s'} \gamma = \sum_s P(s) = 1$), and the objective equals $\mathbb{E}_Q[X]$. The program is feasible (take $\gamma(s, s') = P(s)\mathbf{1}[s' = s]$, of transport cost $0 \leq \varepsilon$) and bounded (X is bounded), so LP strong duality applies. Attaching a free multiplier $u(s)$ to each marginal equality and $\lambda \geq 0$ to the budget, the Lagrangian is

$$\sum_s P(s)u(s) + \lambda\varepsilon + \sum_{s,s'} \gamma(s, s')(X(s') - u(s) - \lambda d(s, s')).$$

Its supremum over $\gamma \geq 0$ is finite iff $X(s') - u(s) - \lambda d(s, s') \leq 0$ for all s, s' , i.e. $u(s) \geq \max_{s'} (X(s') - \lambda d(s, s')) = X_\lambda^c(s)$, in which case the supremum equals $\sum_s P(s)u(s) + \lambda\varepsilon$ (attained at $\gamma = 0$). Minimizing over feasible u sets $u(s) = X_\lambda^c(s)$, giving

$$\text{WEVaR}_\varepsilon(X) = \min_{\lambda \geq 0} \left\{ \lambda\varepsilon + \sum_s P(s) X_\lambda^c(s) \right\} = \inf_{\lambda \geq 0} \left\{ \lambda\varepsilon + \mathbb{E}_P[X_\lambda^c] \right\},$$

which is (4). (For general Polish spaces this is Kantorovich–Rubinstein/Wasserstein-DRO strong duality, Villani, 2009, Gao and Kleywegt, 2023, Blanchet and Murthy, 2019.)

(b) Convexity and $\lambda^* \leq \text{Lip}_d(X)$. For each fixed s , $\lambda \mapsto X_\lambda^c(s) = \max_{s'} (X(s') - \lambda d(s, s'))$ is a pointwise maximum of functions affine in λ , hence convex; therefore $\mathbb{E}_P[X_\lambda^c] = \sum_s P(s)X_\lambda^c(s)$ is convex (nonnegative combination), and $\phi(\lambda) := \lambda\varepsilon + \mathbb{E}_P[X_\lambda^c]$ is convex on $[0, \infty)$, so (4) is a well-posed one-dimensional convex program. Next, if $\lambda \geq \text{Lip}_d(X)$ then for every s, s' , $X(s') - X(s) \leq \text{Lip}_d(X) d(s, s') \leq \lambda d(s, s')$, so $X(s') - \lambda d(s, s') \leq X(s)$ with equality at $s' = s$; hence $X_\lambda^c(s) = X(s)$ and $\phi(\lambda) = \lambda\varepsilon + \mathbb{E}_P[X]$, which is strictly increasing in λ . A convex ϕ that is increasing on $[\text{Lip}_d(X), \infty)$ attains its minimum at some $\lambda^* \leq \text{Lip}_d(X)$.

(c) Monotonicity and concavity in ε . The feasible set $\{Q : \mathcal{W}_1(Q, P) \leq \varepsilon\}$ grows with ε , so $\text{WEVaR}_\varepsilon(X)$ is nondecreasing. By (4) it is an infimum over λ of the maps $\varepsilon \mapsto \lambda\varepsilon + \mathbb{E}_P[X_\lambda^c]$, each affine in ε ; an infimum of affine functions is concave.

(d) Closed form and saturation. Let $T(s) \in \arg \max_{s'} (X(s') - \text{Lip}_d(X) d(s, s'))$ be a steepest-ascent target and set $\bar{\varepsilon}(X) := \mathbb{E}_P[d(s, T(s))] = \sum_s P(s) d(s, T(s))$. The right derivative of the convex function $\mathbb{E}_P[X_\lambda^c]$ at $\lambda = \text{Lip}_d(X)$ equals $-\sum_s P(s) d(s, T(s)) = -\bar{\varepsilon}(X)$ (each active term contributes the negative source–target distance, inactive terms contribute 0). Hence the left derivative of ϕ at $\text{Lip}_d(X)$ is $\varepsilon - \bar{\varepsilon}(X)$, which is ≤ 0 iff $\varepsilon \leq \bar{\varepsilon}(X)$; in that regime the minimizer is $\lambda^* = \text{Lip}_d(X)$ and

$$\text{WEVaR}_\varepsilon(X) = \text{Lip}_d(X) \varepsilon + \mathbb{E}_P[X_{\text{Lip}_d(X)}^c] = \mathbb{E}_P[X] + \varepsilon \text{Lip}_d(X).$$

For $|\text{supp}(P) \cup \{T(s)\}| = 2$ there is a single transport channel, $\bar{\varepsilon}(X) = \mathcal{W}_1(\delta_{\arg \max X}, \cdot)$ exhausts only at a vertex, and the formula is exact up to that point. $\square$

B. PROPOSITION 1 (COHERENCE)

Write $\rho(X) = \text{WEVaR}_\varepsilon(X) = \sup_{Q \in B} \mathbb{E}_Q[X]$ with $B = \{Q : \mathcal{W}_1(Q, P) \leq \varepsilon\}$, a nonempty ($P \in B$), convex, compact subset of $\Delta(\mathcal{S})$. *Monotonicity:* if $X \leq Y$ pointwise then $\mathbb{E}_Q[X] \leq \mathbb{E}_Q[Y]$ for every $Q \in B$ (as $Q \geq 0$), so

$\rho(X) \leq \rho(Y)$. *Translation invariance*: for $c \in \mathbb{R}$, $\mathbb{E}_Q[X + c] = \mathbb{E}_Q[X] + c$ for all Q , hence $\rho(X + c) = \rho(X) + c$. *Positive homogeneity*: for $t \geq 0$, $\rho(tX) = \sup_{Q \in B} t \mathbb{E}_Q[X] = t \rho(X)$. *Subadditivity*: $\rho(X + Y) = \sup_{Q \in B} (\mathbb{E}_Q[X] + \mathbb{E}_Q[Y]) \leq \sup_{Q \in B} \mathbb{E}_Q[X] + \sup_{Q \in B} \mathbb{E}_Q[Y] = \rho(X) + \rho(Y)$. These are the axioms of Artzner et al. [1999]; B is the risk envelope of the induced dual representation. $\square$

C. THEOREM 2

(i) Sweep from mean to worst case. $\mathcal{W}_1$ is a metric on $\Delta(\mathcal{S})$, so $\mathcal{W}_1(Q, P) = 0 \iff Q = P$ and $\text{WEVaR}_0(X) = \mathbb{E}_P[X]$. For any Q , $\mathbb{E}_Q[X] \leq \max_s X(s)$, so $\text{WEVaR}_\varepsilon(X) \leq \max_s X(s)$. Let $s^\bullet \in \arg \max_s X(s)$. Then $\mathcal{W}_1(\delta_{s^\bullet}, P) = \sum_s P(s) d(s, s^\bullet) \leq D$, so $\delta_{s^\bullet} \in B$ once $\varepsilon \geq \sum_s P(s) d(s, s^\bullet)$, whence $\text{WEVaR}_\varepsilon(X) = \max_s X(s)$. Combined with monotonicity (Thm. 1(c)), $\text{WEVaR}_\varepsilon(X) \uparrow \max_s X(s)$ as $\varepsilon \uparrow D$.

Lemma (transport vs. total variation). For all Q, P , $\mathcal{W}_1(Q, P) \leq D \cdot \text{TV}(Q, P)$, where $\text{TV}(Q, P) = \frac{1}{2} \sum_s |Q(s) - P(s)|$. *Proof*: let $m(s) = \min(Q(s), P(s))$; the coupling that keeps mass m in place ($\gamma(s, s) \geq m(s)$) and transports the residual mass $\sum_s (Q(s) - m(s)) = \text{TV}(Q, P)$ arbitrarily incurs cost $\leq D \cdot \text{TV}(Q, P)$, and $\mathcal{W}_1$ is the minimal cost. $\diamond$

(ii) Sandwich. Fix $\alpha \in (0, 1]$ and put $\rho = -\ln \alpha$. For any Q with $\text{KL}(Q\|P) \leq \rho$, Pinsker's inequality gives $\text{TV}(Q, P) \leq \sqrt{\text{KL}(Q\|P)/2} \leq \sqrt{\rho/2}$, so by the Lemma $\mathcal{W}_1(Q, P) \leq D \sqrt{\rho/2} = D \sqrt{-\frac{1}{2} \ln \alpha}$. Therefore $\{Q : \text{KL}(Q\|P) \leq -\ln \alpha\} \subseteq \{Q : \mathcal{W}_1(Q, P) \leq D \sqrt{-\frac{1}{2} \ln \alpha}\}$, and taking the supremum of $\mathbb{E}_Q[X]$ over the larger set,

$$\text{EVaR}_\alpha(X) = \sup_{\text{KL}(Q\|P) \leq -\ln \alpha} \mathbb{E}_Q[X] \leq \sup_{\mathcal{W}_1(Q, P) \leq D \sqrt{-\frac{1}{2} \ln \alpha}} \mathbb{E}_Q[X] = \text{WEVaR}_{D \sqrt{-\frac{1}{2} \ln \alpha}}(X).$$

(iii) Catastrophe domination. Assume $P(s^*) = 0$ and $X(s^*) > \mathbb{E}_P[X]$. *Entropic measure is blind.* $\mathbb{E}_P[e^{tX}] = \sum_{s: P(s) > 0} P(s) e^{tX(s)}$ omits the term s^* , so it—and hence $\text{EVaR}_\alpha(X) = \inf_{t > 0} \frac{1}{t} (\ln \mathbb{E}_P[e^{tX}] - \ln \alpha)$ —does not depend on $X(s^*)$, for every α . (Dually, any Q with $\text{KL}(Q\|P) < \infty$ satisfies $Q \ll P$, so $Q(s^*) = 0$ and $\mathbb{E}_Q[X]$ ignores $X(s^*)$.) *Transport measure reacts.* Let $\delta = \text{dist}_d(s^*, \text{supp } P) = \min_{s: P(s) > 0} d(s, s^*)$ and s_0 an achieving state. For $\varepsilon > \delta$ choose $\eta_0 = \min(P(s_0), \varepsilon/\delta) > 0$ and the feasible law $Q_{\eta_0} = P - \eta_0 \delta_{s_0} + \eta_0 \delta_{s^*}$, for which $\mathcal{W}_1(Q_{\eta_0}, P) \leq \eta_0 \delta \leq \varepsilon$. Then

$$\text{WEVaR}_\varepsilon(X) \geq \mathbb{E}_{Q_{\eta_0}}[X] = \mathbb{E}_P[X] + \eta_0 (X(s^*) - X(s_0)),$$

which is strictly increasing in $X(s^*)$. Moreover $\text{WEVaR}_\varepsilon(X) = \max_{Q \in B} \sum_s Q(s) X(s)$ is convex in the vector X , and by Danskin's theorem its partial right-derivative in $X(s^*)$ equals $\max\{Q(s^*) : Q \text{ optimal}\}$; for $X(s^*)$ large enough any optimal Q must place positive mass on s^* (else Q_{η_0} strictly improves), so the derivative is positive and $\text{WEVaR}_\varepsilon(X)$ strictly increases in $X(s^*)$ whenever $\varepsilon > \delta$. $\square$

D. THEOREM 3

Fix (s, a, b) and write the continuation vector $W(s') = V(s', \psi(b, s, a, s'))$ and radius $\varepsilon(b) = \beta \mathcal{H}(b)$, nominal $\bar{P}_b$.

(i) Closed form of the inner minimization. Applying Theorem 1 to the loss $-W$ and using $(-W)_\lambda^c(s) = \max_{s'} (-W(s') - \lambda d(s', s)) = -\min_{s'} (W(s') + \lambda d(s', s))$,

$$\inf_{Q \in \mathcal{U}(b)} \mathbb{E}_Q[W] = -\text{WEVaR}_{\varepsilon(b)}(-W) = \sup_{\lambda \geq 0} \left\{ \mathbb{E}_{\bar{P}_b} [W_\lambda^{c,-}] - \lambda \varepsilon(b) \right\}, \quad W_\lambda^{c,-}(s) = \min_{s'} (W(s') + \lambda d(s, s')),$$

a one-dimensional concave maximization, with first-order/unsaturated value $\mathbb{E}_{\bar{P}_b}[W] - \beta \mathcal{H}(b) \text{Lip}_d(W)$ (Thm. 1(d) applied to $-W$). Evaluating $\text{Lip}_d(W)$ costs $O(n^2)$ and the dual is a scalar program, versus an $O(n^2)$ -variable transportation LP (6).

(ii) Contraction. We use two properties of $\mathfrak{T}$. *Monotonicity*: if $V \leq V'$ then $W \leq W'$ pointwise, so $\inf_Q \mathbb{E}_Q[W] \leq \inf_Q \mathbb{E}_Q[W']$ and, taking $\max_a$, $\mathfrak{T}V \leq \mathfrak{T}V'$. *Scaled constant shift*: for $c \in \mathbb{R}$, adding c to V adds c to every continuation, and $\inf_Q \mathbb{E}_Q[W + c] = \inf_Q \mathbb{E}_Q[W] + c$, so $\mathfrak{T}(V + c\mathbf{1}) = \mathfrak{T}V + \gamma c\mathbf{1}$. For $\gamma < 1$, monotonicity and the scaled shift are exactly the hypotheses of the contraction lemma [Bertsekas and Tsitsiklis, 1996, Ch. 2]: $\mathfrak{T}V - \mathfrak{T}V' \leq \gamma \|V - V'\|_\infty \mathbf{1}$ and symmetrically, so $\|\mathfrak{T}V - \mathfrak{T}V'\|_\infty \leq \gamma \|V - V'\|_\infty$, a γ -contraction. For $\gamma = 1$ the shift is exact and gives nonexpansiveness;

under properness—every policy together with every kernel selection in the sets $\mathcal{U}(\cdot)$ reaches the terminal set $\mathcal{T}$ in uniformly bounded expected time—the standard stochastic-shortest-path argument [Bertsekas and Tsitsiklis, 1996, Ch. 3] upgrades this to a contraction in a weighted supremum norm $\|\cdot\|_w$: there exist $m \geq 1$, $\kappa \in [0, 1)$ with $\|\mathfrak{T}^m V - \mathfrak{T}^m V'\|_w \leq \kappa \|V - V'\|_w$. Either way $\mathfrak{T}$ has a unique fixed point V^* .

(iii) Safety sandwich. *Upper bound.* Since $\bar{P}_b \in \mathcal{U}(b)$, $\inf_{Q \in \mathcal{U}(b)} \mathbb{E}_Q[W] \leq \mathbb{E}_{\bar{P}_b}[W]$, so $\mathfrak{T}V \leq \mathfrak{T}_{\text{Bayes}}V$ pointwise, where $\mathfrak{T}_{\text{Bayes}}$ is the non-robust operator with kernel $\bar{P}_b = \sum_z b(z)P(\cdot | s, a, z)$; by monotone iteration $V^* \leq V_{\text{Bayes}}$. Committing to one policy under belief b cannot beat knowing the type, so $V_{\text{Bayes}}(s, b) \leq \mathbb{E}_{z \sim b}[V_{\text{opt}}^*(s, z)]$ (nonnegative value of information), giving the upper bound. *Lower bound.* Let V_{wc}^* be the fixed point of the operator $\mathfrak{T}_{\text{wc}}$ obtained by replacing the radius $\beta\mathcal{H}(b)$ with its maximum $\beta \ln |\mathcal{Z}|$ at every (s, b) , keeping the same nominal $\bar{P}_b$. Since $\beta\mathcal{H}(b) \leq \beta \ln |\mathcal{Z}|$, the ambiguity ball in $\mathfrak{T}$ is contained in that of $\mathfrak{T}_{\text{wc}}$, so its inner infimum is no smaller; hence $\mathfrak{T}V \geq \mathfrak{T}_{\text{wc}}V$ pointwise and, by monotone iteration, $V^*(s, b) \geq V_{\text{wc}}^*(s, b)$. This is the bound used in Fig. 3 and requires no nested-set condition. *Convergence.* As $b \rightarrow \delta_{z^*}$, $\mathcal{H}(b) \rightarrow 0$ and $\varepsilon(b) \rightarrow 0$, so $\mathcal{U}(b) \rightarrow \{P(\cdot | s, a, z^*)\}$; by the Lipschitz dependence of the value on the radius (Thm. 1(c): $0 \leq \text{WEVaR}_\varepsilon - \text{WEVaR}_0 \leq \varepsilon \text{Lip}_d$), $V^*(s, b) \rightarrow V_{\text{opt}}^*(s, z^*)$. $\square$

E. SUPPLEMENTARY EXPERIMENTS AND FIGURES

All figures are reproduced by the released code (a Colab-ready package). The body shows the geometric core (Figs. 1, 2) and the certified guarantee (Fig. 3); the remaining plots are collected here.

Quantitative verification of the dualities. Figure 4 is the solver-side companion to §4: across a sweep of radii the entropic and Wasserstein measures both rise from the mean to the worst case, with $\text{EVaR}_\alpha \leq \text{WEVaR}$ at the Pinsker-matched radius (Thm. 2(ii)); and a zero-nominal-probability disaster leaves the entropic measure flat while WEVaR reacts (Thm. 2(iii)).

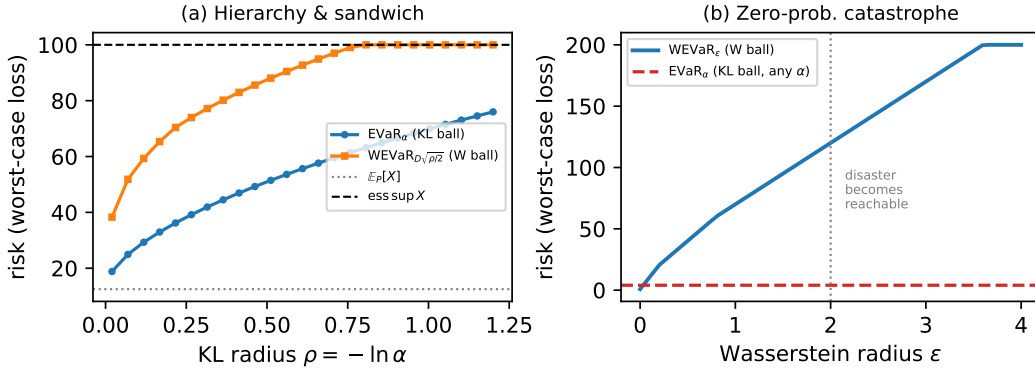


Figure 4: Gurobi-verified comparison on a five-state metric space. **(a)** hierarchy and the Pinsker sandwich; **(b)** the zero-probability catastrophe: the entropic value-at-risk is invariant to the disaster’s magnitude, WEVaR is not.

Manifolds over the belief simplex. For a three-type environment the belief lives on a 2-simplex, and the controller reads three scalar fields off it (Fig. 5): the radius $\varepsilon(b) = \beta\mathcal{H}(b)$ (an entropy bowl, maximal at the centroid and zero at the vertices), the robust value $V^*(s_0, b)$, and the policy regions.

Safety switches. The belief-dependent policy flips from the safe to the optimal action at a computable threshold (Fig. 6): on the Ambiguous Bridge at $\alpha^* \approx 0.983$ (set by the reward asymmetry, independent of β), and on the stormy-drone canyon at $b(\text{calm}) \approx 0.63$.

Safety-vs-efficiency rollouts, and an honest caveat. Figure 7 compares the three planners on the canyon. WEVaR Pareto-dominates the static-robust (MAXIMIN) planner—comparable safety at higher return, the “un-freezing” effect. In this *calibrated-belief* setting the expected-value (Bayesian) planner is already near-optimally cautious under a severe catastrophe, so the empirical separation between WEVaR and Bayesian is small; the regime in which belief-scaled caution measurably reduces failures is one of *miscalibrated* belief or distribution shift, where the belief is learned rather than computed from a known model. The body’s contribution is therefore the coherent measure, its geometry, and the certified guarantee; this rollout is illustrative.

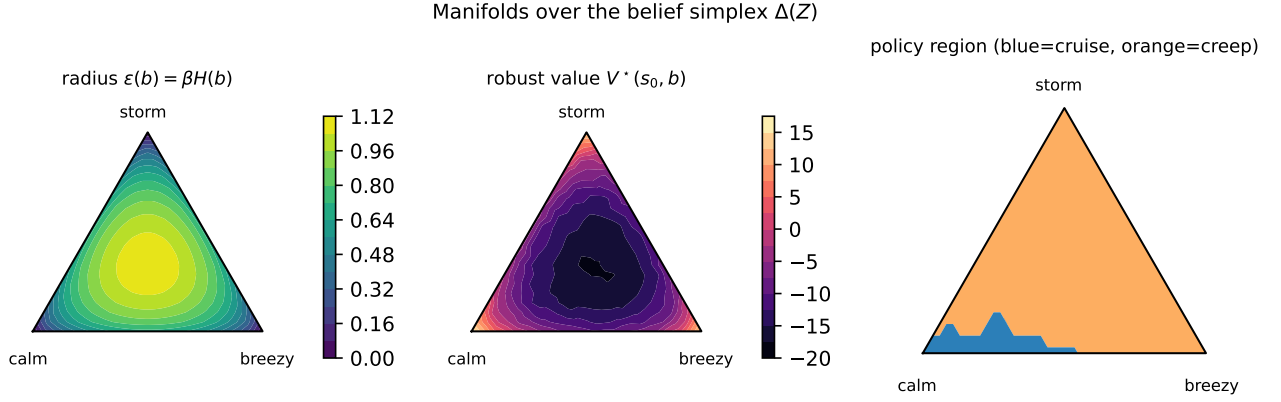


Figure 5: Manifolds over the belief simplex $\Delta(\mathcal{Z})$, $|\mathcal{Z}| = 3$ (calm/breezy/storm): ambiguity radius, robust value, and policy region (cruise only near the calm vertex).

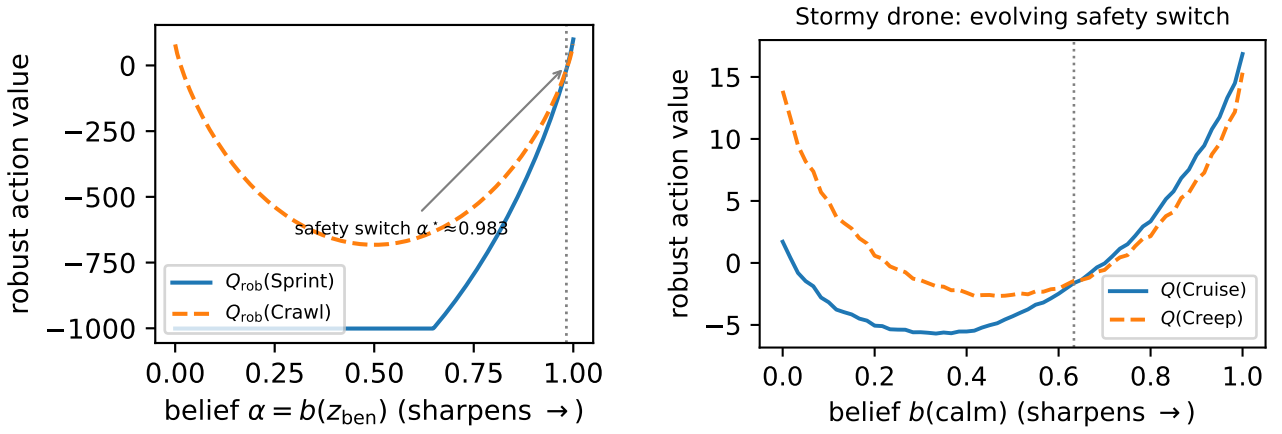


Figure 6: Robust action values vs. belief. Left: the Ambiguous Bridge (SPRINT vs. CRAWL, switch at $\alpha^* \approx 0.983$). Right: the stormy-drone canyon (CRUISE vs. CREEP, switch at $b(\text{calm}) \approx 0.63$).

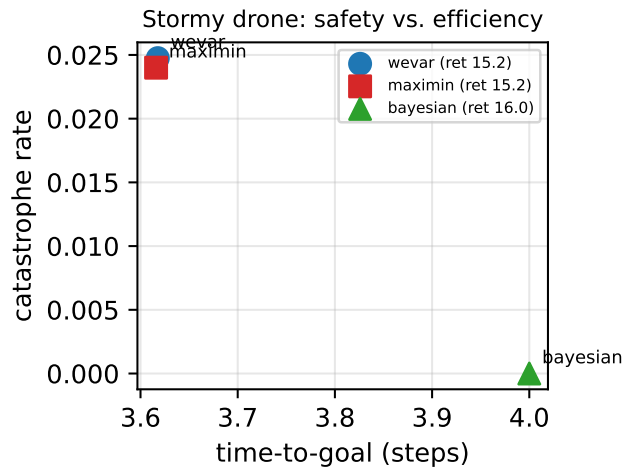


Figure 7: Catastrophe rate vs. time-to-goal on the stormy-drone canyon (4000 episodes). WEVaR dominates the static-robust MAXIMIN baseline; the Bayesian planner is already cautious here (see caveat).